

Linux Commands End to End Cheat Sheet

Command: `whoami`

```
sunith@sunith-virtual-machine:~$ whoami
sunith
sunith@sunith-virtual-machine:~$
```

- Description: Displays the current user account.
- Example: Running `whoami` in the terminal will show your username, like `john_doe`.

Command: `ls`

```
sunith@sunith-virtual-machine:~$ ls
Desktop  Direct1  Direct2  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
sunith@sunith-virtual-machine:~$
```

- Description: Lists files and directories in the current directory.
- Example: `ls` will display files and folders in your current directory, and `ls -l` provides detailed information.

Command: `ip a`

```
sunith@sunith-virtual-machine:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:a7:b1:13 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.181.128/24 brd 192.168.181.255 scope global dynamic noprefixroute ens33
        valid_lft 1260sec preferred_lft 1260sec
    inet6 fe80::76ba:2b11:79f8:c8d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
sunith@sunith-virtual-machine:~$
```

- Description: Lists your IP address and network interfaces.
- Example: `ip a` show all network interfaces and their IP addresses.

Command: `cat`

```
sunith@sunith-virtual-machine:~$ cat myfile.txt
it will show the contents of file
sunith@sunith-virtual-machine:~$
```

- Description: Reads the contents of text files.
- Example: `cat myfile.txt` will display the content of `myfile.txt`.

Command: `passwd`

```
sunith@sunith-virtual-machine:~$ passwd
Changing password for sunith.
Current password:
New password:
```

- Description: Used to change the password of the user account.
- Example: Typing `passwd` and following the prompts will allow you to change your password.

Command: `touch`

```
sunith@sunith-virtual-machine:~$ touch file1
sunith@sunith-virtual-machine:~$ ls
Desktop  Direct1  Direct2  Documents  Downloads  file1  Music  myfile.txt  Pictures  Public  snap  Templates  Videos
sunith@sunith-virtual-machine:~$
```

- Description: Creates empty files, mainly useful to test write ability.
- Example: `touch newfile.txt` creates an empty file named `newfile.txt`.

Command: `pwd`

```
sunith@sunith-virtual-machine:~$ pwd
/home/sunith
sunith@sunith-virtual-machine:~$
```

- Description: Stands for Print Working Directory; shows which directory you are in.
- Example: Running `pwd` might return `/home/john_doe`.

Command: `man`

```
ECHO(1) User Commands

NAME
    echo - display a line of text

SYNOPSIS
    echo [SHORT-OPTION...]... [STRING]...
    echo LONG-OPTION

DESCRIPTION
    Echo the STRING(s) to standard output.

    -n      do not output the trailing newline

    -e      enable interpretation of backslash escapes

    -E      disable interpretation of backslash escapes (default)

    --help  display this help and exit

    --version
            output version information and exit

    If -e is in effect, the following sequences are recognized:

    \\      backslash
    \a      alert (BEL)
```

- Description: Accesses the system documentation for commands.
- Example: `man ls` will show the manual for the `ls` command.

Command: apropos

```
sunith@sunith-virtual-machine:~$ apropos file
00-upstream-settings (5) - dconf configuration file
FILE (3) - overview of system data types
[ (1) - check file types and compare values
aa-exec (1) - confine a program with the specified AppArmor profile
aa-remove-unknown (8) - remove unknown AppArmor profiles
aa-teardown (8) - unload all AppArmor profiles
access.conf (5) - the login access control table file
acct (5) - process accounting file
adduser.conf (5) - configuration file for adduser(8) and addgroup(8) .
anacrontab (5) - configuration file for anacron
aplaymidi (1) - play Standard MIDI Files
apparmor.d (5) - syntax of security profiles for AppArmor.
apparmor.vim (5) - vim syntax highlighting file for AppArmor profiles
apparmor_parser (8) - loads AppArmor profiles into the kernel
apparmor_xattrs (7) - AppArmor profile xattr(7) matching
apport-bug (1) - file a bug report using Apport, or update an existing report
apport-collect (1) - file a bug report using Apport, or update an existing report
apport-unpack (1) - extract the fields of a problem report to separate files
apt-ftpparchive (1) - Utility to generate index files
apt-sortpkgs (1) - Utility to sort package index files
apt.conf (5) - Configuration file for APT
apt_auth.conf (5) - Login configuration file for APT sources and proxies
apt_preferences (5) - Preference control file for APT
arecordmidi (1) - record Standard MIDI Files
aspell-autobuildhash (8) - Autobuilding aspell hash files for some dicts
avahi-daemon.conf (5) - avahi-daemon configuration file
```

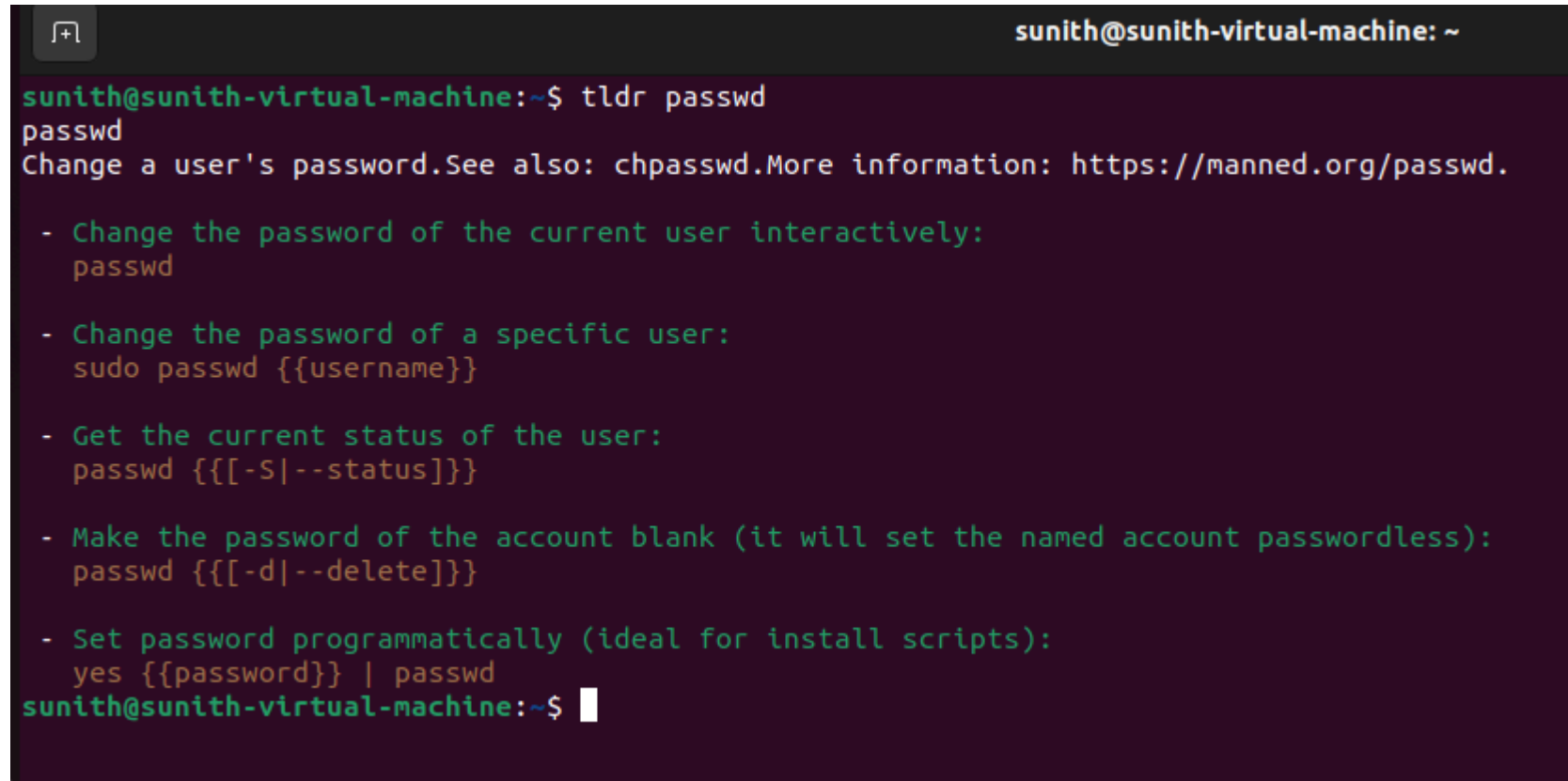
- Description: Searches commands related to a specific task based on description (equivalent to `man -k`).
- Example: `apropos "list files"` might return commands related to listing files.

Command: `pinfo`

```
sunith@sunith-virtual-machine: ~  
File: dir, Node: Top This is the top of the INFO tree  
  
This (the Directory node) gives a menu of major topics.  
Typing "q" exits, "H" lists all Info commands, "d" returns here,  
"h" gives a primer for first-timers,  
"mEmacs<Return>" visits the Emacs manual, etc.  
  
In Emacs, you can click mouse button 2 on a menu item or cross reference  
to select it.  
  
* Menu:  
  
Basics  
* Common options: (coreutils)Common options.  
* Coreutils: (coreutils). Core GNU (file, text, shell) utilities.  
* Date input formats: (coreutils)Date input formats.  
* Ed: (ed). The GNU line editor  
* File permissions: (coreutils)File permissions.  
Access modes.  
* Finding files: (find). Operating on files matching certain criteria.  
* Time: (time). time  
  
Compression  
* Gzip: (gzip). General (de)compression of files (lzw).  
  
Development  
* SSIP: (ssip). Speech Synthesis Interface Protocol.  
* Speech Dispatcher: (speech-dispatcher).  
Viewing line 28/210, 13%  
your computer, move the mouse pointer outside or press Ctrl+Alt.
```

- Description: Provides additional information, similar to `man`.
- Example: `pinfo ls` gives detailed information about the `ls` command.

Command: `tldr`

A terminal window with a dark purple background. The title bar at the top right says "sunith@sunith-virtual-machine: ~". The prompt is "sunith@sunith-virtual-machine:~\$". The command "tldr passwd" has been entered. The output shows the command "passwd" followed by a description: "Change a user's password. See also: chpasswd. More information: https://manned.org/passwd." Below this is a list of five bullet points, each with a command example. The prompt "sunith@sunith-virtual-machine:~\$" is visible at the bottom with a cursor.

```
sunith@sunith-virtual-machine:~$ tldr passwd
passwd
Change a user's password. See also: chpasswd. More information: https://manned.org/passwd.

- Change the password of the current user interactively:
  passwd

- Change the password of a specific user:
  sudo passwd {{username}}

- Get the current status of the user:
  passwd {[ -S | --status ]}

- Make the password of the account blank (it will set the named account passwordless):
  passwd {[ -d | --delete ]}

- Set password programmatically (ideal for install scripts):
  yes {{password}} | passwd
sunith@sunith-virtual-machine:~$
```

- Description: Shows command line examples, a simplified version of `man`.
- Example: `tldr passwd` provides quick examples of how to use the `passwd`

Essential File Management Tools Commands

Command: `ls`

```
sunith@sunith-virtual-machine:~$ ls
Desktop  Direct1  Direct2  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
sunith@sunith-virtual-machine:~$
```

- Example: `ls /home/user/Documents`
- Description: Lists files and directories within a specified path. This command is used to view the contents of directories.

Command: `cp`

```
sunith@sunith-virtual-machine:~$ ls
Desktop  Direct1  Direct2  Documents  Downloads  file1  Music  myfile.txt  Pictures  Public  snap  Templates  Videos
sunith@sunith-virtual-machine:~$ cp file1 /tmp/
sunith@sunith-virtual-machine:~$ ls /tmp/
file1
file2
snap-private-tmp
systemd-private-62c3caeb647a406e861c4d9f493dde6a-chrony.service-B8yYag
systemd-private-62c3caeb647a406e861c4d9f493dde6a-colord.service-IwxluN
systemd-private-62c3caeb647a406e861c4d9f493dde6a-ModemManager.service-YlDA5p
systemd-private-62c3caeb647a406e861c4d9f493dde6a-power-profiles-daemon.service-AXlKTU
systemd-private-62c3caeb647a406e861c4d9f493dde6a-switcheroo-control.service-VJ1fRB
systemd-private-62c3caeb647a406e861c4d9f493dde6a-systemd-logind.service-Uxh3P9
systemd-private-62c3caeb647a406e861c4d9f493dde6a-systemd-oemd.service-obaRBM
systemd-private-62c3caeb647a406e861c4d9f493dde6a-systemd-resolved.service-zvpXp6
systemd-private-62c3caeb647a406e861c4d9f493dde6a-ubuntu-advantage-desktop-daemon.service-cKLLjV
systemd-private-62c3caeb647a406e861c4d9f493dde6a-upower.service-dBBPyA
tmpmpvvc5kj
VMwareDnD
vmware-root_768-2999133183
sunith@sunith-virtual-machine:~$
```

- Example: `cp source.txt destination.txt`
- Description: Copies files from one location to another. This command is used to make duplicates of files.

Command: `mkdir`

```
es  Terminal Aug 12 17:51
sunith@sunith-virtual-machine: ~
sunith@sunith-virtual-machine:~$ mkdir sunith
sunith@sunith-virtual-machine:~$ ls
Desktop  Direct1  Direct2  Documents  Downloads  file1  Music  myfile.txt  Pictures  Public  snap  sunith  Templates  Videos
sunith@sunith-virtual-machine:~$
```

- Example: `mkdir new_directory`
- Description: Creates a new directory. This command is used when you need to make a new folder.

Command: `rmdir`

```
sunith@sunith-virtual-machine:~$ mkdir sunith
sunith@sunith-virtual-machine:~$ ls
Desktop  Direct1  Direct2  Documents  Downloads  file1  Music  myfile.txt  Pictures  Public  snap  sunith  Templates  Videos
sunith@sunith-virtual-machine:~$ rmdir sunith
sunith@sunith-virtual-machine:~$
```

- Example: `rmdir empty_directory`
- Description: Removes a directory, but only if it's empty. This command is used for deleting empty folders.

Command: `mv`

```
sunith@sunith-virtual-machine:~$ ls
Desktop  Direct1  Direct2  docfile.txt  Documents  Downloads  file1  Music  Pictures  Public  snap  Templates  Videos
sunith@sunith-virtual-machine:~$
```

- Example: `mv oldname.txt newname.txt`
- Description: Moves or renames files and directories. This command can change the location or the name of a file.

Command: `rm`

```
sunith@sunith-virtual-machine:~$ rm docfile.txt
sunith@sunith-virtual-machine:~$ ls
Desktop  Direct1  Direct2  Documents  Downloads  file1  Music  Pictures  Public  snap  Templates  Videos
sunith@sunith-virtual-machine:~$
```

- Example: `rm filename.txt`
- Description: Removes files or directories. This command deletes files or directories.
- Note: When combined with options `-r` (recursive) and `-f` (force), as in `rm -rf directory_name`, it becomes a powerful command that can delete directories and their contents without asking for confirmation.

Advanced File Management Tools Commands

ln: Used for creating hard and symbolic links.

```
sunith@sunith-virtual-machine:~$ ls
Orginal.txt
sunith@sunith-virtual-machine:~$ ln Orginal.txt Sunith
sunith@sunith-virtual-machine:~$ ls
Orginal.txt  Sunith
sunith@sunith-virtual-machine:~$ ln Orginal.txt Devops.txt
sunith@sunith-virtual-machine:~$ ls
Devops.txt  Orginal.txt  Sunith
sunith@sunith-virtual-machine:~$
```

- Example: `ln -s source_file symbolic_link` creates a symbolic link to `source_file`.
- Description: Links a file or directory to another location.

find: A utility to find files based on various properties.

```
sunith@sunith-virtual-machine:~$ find -name Orginal.txt
./Orginal.txt
sunith@sunith-virtual-machine:~$
```

- Example: `find /home -name "*.txt"` searches for text files in the `/home` directory.
- Description: Searches for files and directories in a given path.

locate: Used for quickly searching the database of system's contents.

```
sunith@sunith-virtual-machine:~$ locate file1
/home/sunith/file1
/usr/share/doc/libsndfile1
/usr/share/doc/libxkbfile1
/usr/share/doc/libsndfile1/changelog.Debian.gz
/usr/share/doc/libsndfile1/copyright
/usr/share/doc/libxkbfile1/changelog.Debian.gz
/usr/share/doc/libxkbfile1/copyright
/var/cache/apt/archives/libsndfile1_1.0.31-2ubuntu0.2_amd64.deb
/var/lib/dpkg/info/libsndfile1:amd64.list
/var/lib/dpkg/info/libsndfile1:amd64.md5sums
/var/lib/dpkg/info/libsndfile1:amd64.shlibs
/var/lib/dpkg/info/libsndfile1:amd64.symbols
/var/lib/dpkg/info/libsndfile1:amd64.triggers
/var/lib/dpkg/info/libxkbfile1:amd64.list
/var/lib/dpkg/info/libxkbfile1:amd64.md5sums
/var/lib/dpkg/info/libxkbfile1:amd64.shlibs
/var/lib/dpkg/info/libxkbfile1:amd64.triggers
sunith@sunith-virtual-machine:~$
```

- Example: `locate filename` finds the location of `filename`.
- Description: Fast file searching, requires running `updatedb` for an updated database.

which: Finds the location of executables in the PATH.

```
sunith@sunith-virtual-machine:~$ which man
/usr/bin/man
sunith@sunith-virtual-machine:~$
```

- Example: `which python` shows the path of the Python executable.
- Description: Useful for finding executable files in system's PATH.

tar: Archives multiple files into a single file.

```
sunith@sunith-virtual-machine:~$ ls
Devops.txt  Orginal.txt  Sunith
sunith@sunith-virtual-machine:~$ which man
/usr/bin/man
sunith@sunith-virtual-machine:~$ ls
Devops.txt  Orginal.txt  Sunith
sunith@sunith-virtual-machine:~$ tar -czvf archive.tar.gz Orginal.txt
Orginal.txt
sunith@sunith-virtual-machine:~$ ls
archive.tar.gz  Devops.txt  Orginal.txt  Sunith
sunith@sunith-virtual-machine:~$
```

- Example: `tar -cvf archive.tar /folder` creates an archive of a folder.
- Description: Combines multiple files into a tarball.

gzip: Compresses files.

```
sunith@sunith-virtual-machine:~$ gzip techie.txt
sunith@sunith-virtual-machine:~$ ls
archive.tar  Devops.txt  Orginal.txt  Sunith  techie.txt.gz
sunith@sunith-virtual-machine:~$
```

- Example: `gzip file.txt` compresses `file.txt`.
- Description: Reduces the size of files using compression.

bzip2: A file compression utility.

```
sunith@sunith-virtual-machine:~$ bzip2 Techie_Horizon.txt
sunith@sunith-virtual-machine:~$ ls
archive.tar  Devops.txt  Orginal.txt  Sunith  Techie_Horizon.txt.bz2  techie.txt.gz
sunith@sunith-virtual-machine:~$
```

- Example: `bzip2 file.txt` compresses `file.txt`.
- Description: Offers better compression ratios than `gzip` at the cost of speed.

xz: High compression utility

```
sunith@sunith-virtual-machine:~$ xz File1
sunith@sunith-virtual-machine:~$ ls
File1.xz  File2  File3  File4  File5  File6
sunith@sunith-virtual-machine:~$
```

- Example: `xz file.txt` compresses `file.txt`.
- Description: Provides the best compression ratio.

zip:Used for packaging and compressing (archiving) files.

```
sunith@sunith-virtual-machine:~$ ls
File1.xz  File2  File3  File4  File5  File6
sunith@sunith-virtual-machine:~$ zip archive.zip File2 File3
  adding: File2 (stored 0%)
  adding: File3 (stored 0%)
sunith@sunith-virtual-machine:~$ ls
archive.zip  File1.xz  File2  File3  File4  File5  File6
sunith@sunith-virtual-machine:~$
```

- Example: `zip archive.zip file1 file2` creates an archive with file1 and file2.
- Description: Commonly used for creating .zip files.

file: Determines the type of a file.

```
sunith@sunith-virtual-machine:~$ file archive.zip
archive.zip: Zip archive data, at least v1.0 to extract, compression method=store
sunith@sunith-virtual-machine:~$ file File1.xz
File1.xz: XZ compressed data, checksum CRC64
sunith@sunith-virtual-machine:~$
```

- Example: `file image.jpg` displays the type of image.jpg.
- Description: Useful for identifying file formats.

mount: Attaches a file system to a specified directory.

```
sunith@sunith-virtual-machine:~$ echo "Hello from Sunith file" > Sunith.txt
sunith@sunith-virtual-machine:~$ cat Sunith.txt
Hello from Sunith file
sunith@sunith-virtual-machine:~$ touch chintu.txt
sunith@sunith-virtual-machine:~$ ls
archive.zip  chintu.txt  File1.xz  File2  File3  File4  File5  File6  Sunith.txt
sunith@sunith-virtual-machine:~$ sudo mount --bind Sunith.txt chintu.txt
[sudo] password for sunith:
sunith@sunith-virtual-machine:~$ ls
archive.zip  chintu.txt  File1.xz  File2  File3  File4  File5  File6  Sunith.txt
sunith@sunith-virtual-machine:~$ cat chintu.txt
Hello from Sunith file
sunith@sunith-virtual-machine:~$
```

- Example: `mount /dev/sdb1 /mnt` mount the device `sdb1` to `/mnt`.
- Description: Used to access file systems on different storage devices.

df: Displays disk space usage of all mounted file systems.

```
sunith@sunith-virtual-machine:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           288M  2.1M  286M   1% /run
/dev/sda3        20G   13G   5.5G  70% /
tmpfs           1.5G     0   1.5G   0% /dev/shm
tmpfs           5.0M  4.0K   5.0M   1% /run/lock
/dev/sda2        512M  6.1M  506M   2% /boot/efi
tmpfs           288M  168K  288M   1% /run/user/1000
/dev/sr0         155M  155M     0 100% /media/sunith/CDROM
sunith@sunith-virtual-machine:~$
```

- Example: `df -h` shows disk space in human-readable format.

- Description: Useful for monitoring disk space usage.

findmnt: Lists all mounted filesystems.

```

- /snap/bare/5 /dev/loop0 squashfs ro,nodev,relatime,errors=continue,threads=single
- /snap/snapd/21759 /dev/loop3 squashfs ro,nodev,relatime,errors=continue,threads=single
- /snap/core22/1612 /dev/loop1 squashfs ro,nodev,relatime,errors=continue,threads=single
- /snap/snap-store/1113 /dev/loop4 squashfs ro,nodev,relatime,errors=continue,threads=single
- /snap/snapd-desktop-integration/178 /dev/loop5 squashfs ro,nodev,relatime,errors=continue,threads=single
- /snap/firefox/4848 /dev/loop2 squashfs ro,nodev,relatime,errors=continue,threads=single
- /snap/gtk-common-themes/1535 /dev/loop6 squashfs ro,nodev,relatime,errors=continue,threads=single
- /snap/gnome-42-2204/176 /dev/loop7 squashfs ro,nodev,relatime,errors=continue,threads=single
- /boot/efi /dev/sda2 vfat rw,relatime,fmask=0077,dmask=0077,codepage=437,iocharset=iso
- /home/sunith/chintu.txt /dev/sda3[/home/sunith/Sunith.txt]
- /media/sunith/CDROM /dev/sr0 ext4 rw,relatime,errors=remount-ro
iso9660 ro,nosuid,nodev,relatime,nojoliet,check=s,map=n,blocksize=20

```

- Example: `findmnt -text4` lists all ext4 type mounts.
- Description: Displays a tree-like structure of mounts.

lsblk: Lists information about block devices.

```
sunith@sunith-virtual-machine:~$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
fd0          2:0    1     4K  0 disk
loop0        7:0    0     4K  1 loop /snap/bare/5
loop1        7:1    0  74.3M  1 loop /snap/core22/1612
loop2        7:2    0 271.2M  1 loop /snap/firefox/4848
loop3        7:3    0  38.8M  1 loop /snap/snapd/21759
loop4        7:4    0  12.9M  1 loop /snap/snap-store/1113
loop5        7:5    0   500K  1 loop /snap/snapd-desktop-integration/178
loop6        7:6    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop7        7:7    0 505.1M  1 loop /snap/gnome-42-2204/176
sda          8:0    0    20G  0 disk
├─sda1       8:1    0     1M  0 part
├─sda2       8:2    0   513M  0 part /boot/efi
└─sda3       8:3    0  19.5G  0 part /home/sunith/chintu.txt
sr0         11:0    1 154.7M  0 rom  /media/sunith/CDROM
sr1         11:1    1 1024M  0 rom
```

- Example: `lsblk` lists all block devices.
- Description: Useful for identifying storage devices and their partitions